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<https://swayam.gov.in>https://swayam.gov.in/nc_details/NPTEL

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NPTEL (<https://swayam.gov.in/explorer?ncCode=NPTEL>) » Introduction to Graph Algorithms (course)

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Course outline

About
NPTEL ()

How does an
NPTEL
online
course
work? ()

Week 1
Introduction
and
Principles of
Algorithms ()

Week 2
Shortest
Path
Algorithms
and
problems ()

Week 3
Bellman
Equations ()

Week 3 Assignment 3

The due date for submitting this assignment has passed.

Due on 2025-08-13, 23:59 IST.

As per our records you have not submitted this assignment.

1) State True or False

1 point

The Bellman equation is not solvable when G has only positive and zero cycles.

- ☐ True
☐ False

No, the answer is incorrect.
Score: 0

Accepted Answers:
False

2) State True or False

1 point

The Bellman equation may not have unique solution when G has zero cycles.

- ☐ True
☐ False

No, the answer is incorrect.
Score: 0

Accepted Answers:
True

3) State True or False

1 point

The Bellman equation do not have a solution if G has a negative cycle.

- ☐ True
☐ False

No, the answer is incorrect.

☐ Belman Equation (unit? unit=20&lesso n=46)

☐ Shortest Path Trees (unit? unit=20&lesso n=47)

☐ Bounded Walks and Paths (unit? unit=20&lesso n=48)

☐ Bounded Walks and Paths 2 (unit? unit=20&lesso n=49)

☐ Week 3 Lecture Material (unit? unit=20&lesso n=64)

☐ Quiz: Week 3 Assignment 3 (assessment? name=99)

☐ Feedback Form (unit? unit=20&lesso n=28)

Week 4
Bellman Ford and Dijkstra Algorithm ()

Week 5 All
Pair Shortest Path ()

Week 6
Prims and Kruskal's Algorithm ()

Week 7 DFS
and Cut Vertex ()

Score: 0

Accepted Answers:

True

4) State True or False

The solution of a Bellman equation is unique if G has no negative cycle or zero cycle.

☐ True

☐ False

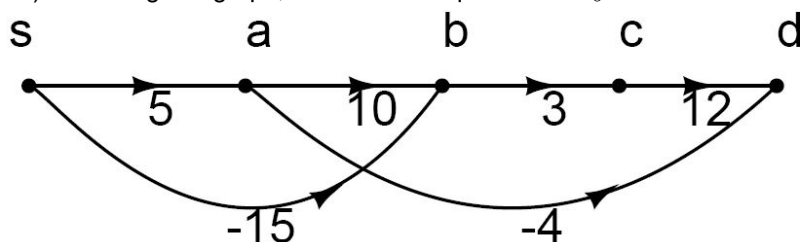
No, the answer is incorrect.

Score: 0

Accepted Answers:

True

5) For the given graph, the Bellman Equation for $x_s =$



☐ 0

☐ $-15 + x_b$

☐ $5 + x_a$

☐ Undefined

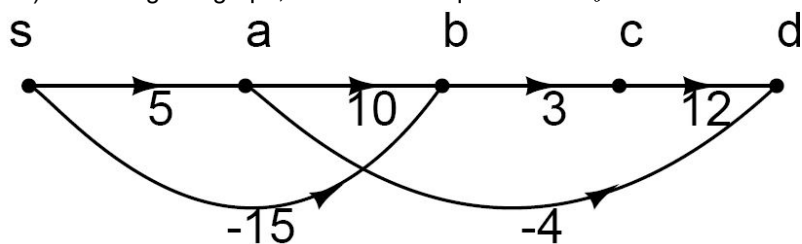
No, the answer is incorrect.

Score: 0

Accepted Answers:

0

6) For the given graph, the Bellman Equation for $x_b =$



☐ $x_a + 10$

☐ $x_s - 15$

☐ 15

☐ $\text{Min}\{x_a + 10, x_s - 15\}$

No, the answer is incorrect.

Score: 0

Accepted Answers:

$\text{Min}\{x_a + 10, x_s - 15\}$

1 point

1 point

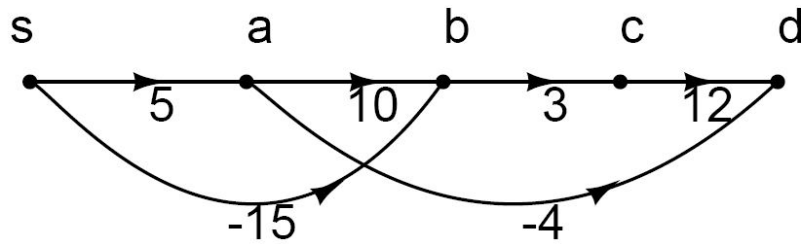
1 point

Video
download ()

Live session
()

7) For the given graph, the Bellman Equation for $x_c =$

1 point



- ☐ $x_a + 13$
☐ $x_b + 3$
☐ $x_s + 18$
☐ $\text{Min}\{x_a + 13, x_b + 3\}$

No, the answer is incorrect.

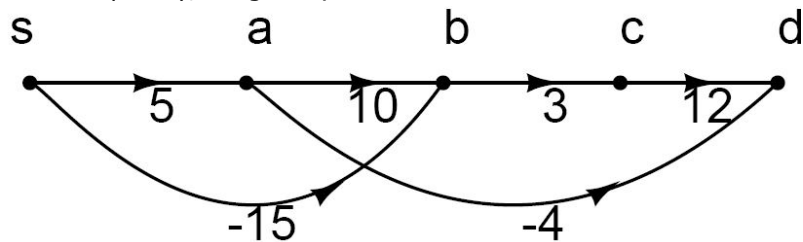
Score: 0

Accepted Answers:

$x_b + 3$

8) If we formulate all the Bellman Equation for the given graph, and solve completely, we get $x_b =$

1 point



- ☐ 15
☐ -15
☐ 0
☐ undefined

No, the answer is incorrect.

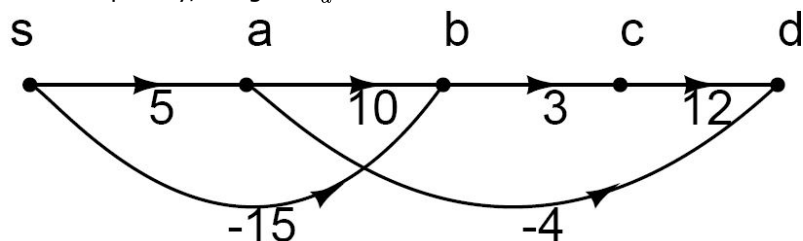
Score: 0

Accepted Answers:

-15

9) If we formulate all the Bellman Equation for the given graph, and solve completely, we get $x_a =$

1 point



- ☐ 5
☐ 0
☐ -5
☐ undefined

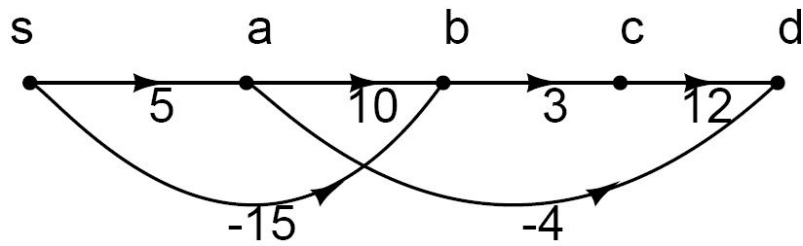
No, the answer is incorrect.

Score: 0

Accepted Answers:

5

10) If we formulate all the Bellman Equation for the given graph, and solve completely, we get $x_d =$



☐ 30

☐ 1

☐ 0

☐ undefined

No, the answer is incorrect.

Score: 0

Accepted Answers:

0

